# DBMS Lab Report

**<u>Team Details: -</u>**
**Team Name: Team ROZ**
**Member:**

- **Ahnaf Wadud Arnab**
  **ID : 011 222 126**
  **Role: Frontend, Backend, Database Designer & Tester.**

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**<u>Introduction: -</u>**

**Project Name: Railway Systems App**

The Railway Systems App is a comprehensive platform designed to simplify booking, managing, and tracking train journeys. It provides a seamless and efficient travel experience through a user-friendly interface and advanced features.

The app includes:

- Real-time train tracking for live location updates
- Detailed route information for precise trip planning
- Simplified round-trip bookings for added convenience
- Boarding point navigation with live location assistance
- Integrated emergency alert services for passenger safety

By addressing common travel challenges, this system enhances convenience, safety, and efficiency, making train travel hassle-free for both daily commuters and long-distance travelers.

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**<u>Links</u>**

- GitHub Repository: [https://github.com/AhnafwadudArnab/Amar-Rail-sheba.git]
- YouTube Video: [https://youtu.be/fCflNXo_EEM]
- Google Drive link: [https://drive.google.com/drive/folders/1pwulqHzOKZpr3uVD8P5x_M5Tsi3ZmyAj?usp=sharing]

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**Motivation**

Train Travel is an essential mode of transportation, yet passengers frequently encounter inefficiencies such as difficulty in booking round trips, lack of real-time updates, and security concerns. Our app bridges these gaps by integrating modern technological solutions, offering an efficient, transparent, and secure railway management system.

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**Key Features**

1. User-Friendly Booking System – Seamless ticket reservation process.
2. Train Filtering – Search trains by class, time, and fare.
3. Real-Time Availability Updates – Check train seat availability instantly.
4. Live Train Location Tracking – Stay updated with real-time train movements.
5. Lost and Found System – Report and recover lost belongings.
6. Round-Trip Booking – Book return journeys in a single transaction.
7. Emergency Alerts for Passengers – Notify authorities in critical situations.

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**Development Stack**

- Frontend: Flutter (Dart)
- Backend: Node.js, Express.js
- Database: MySQL
- IDE: Android Studio

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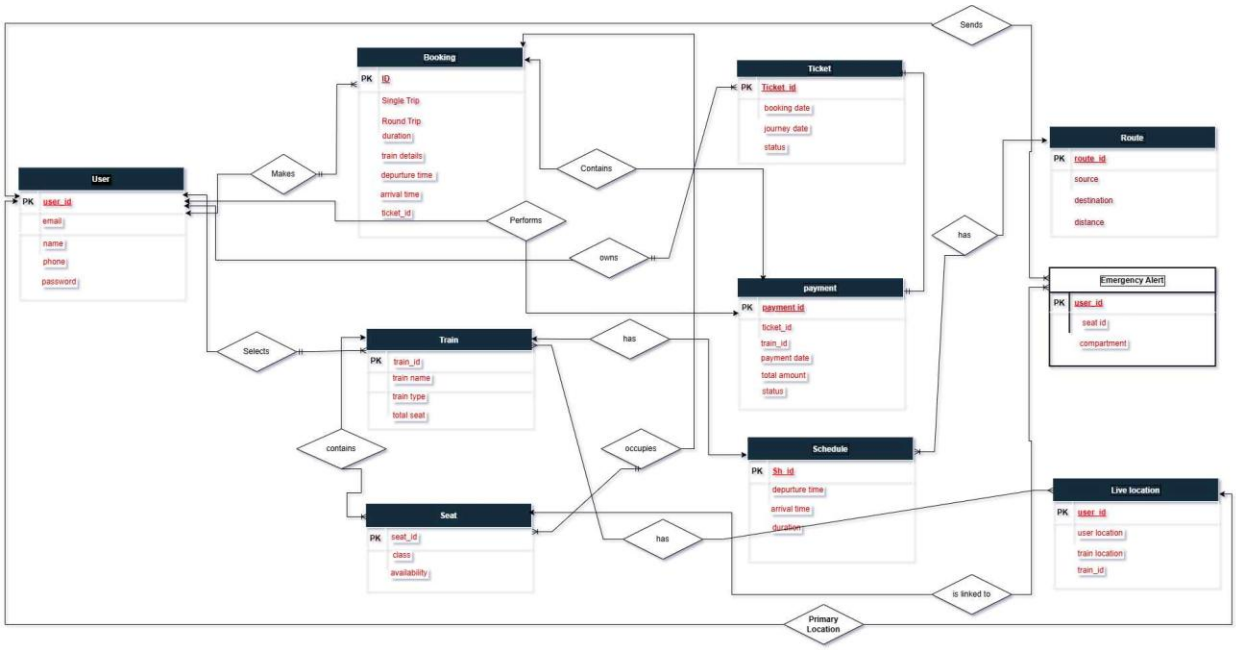
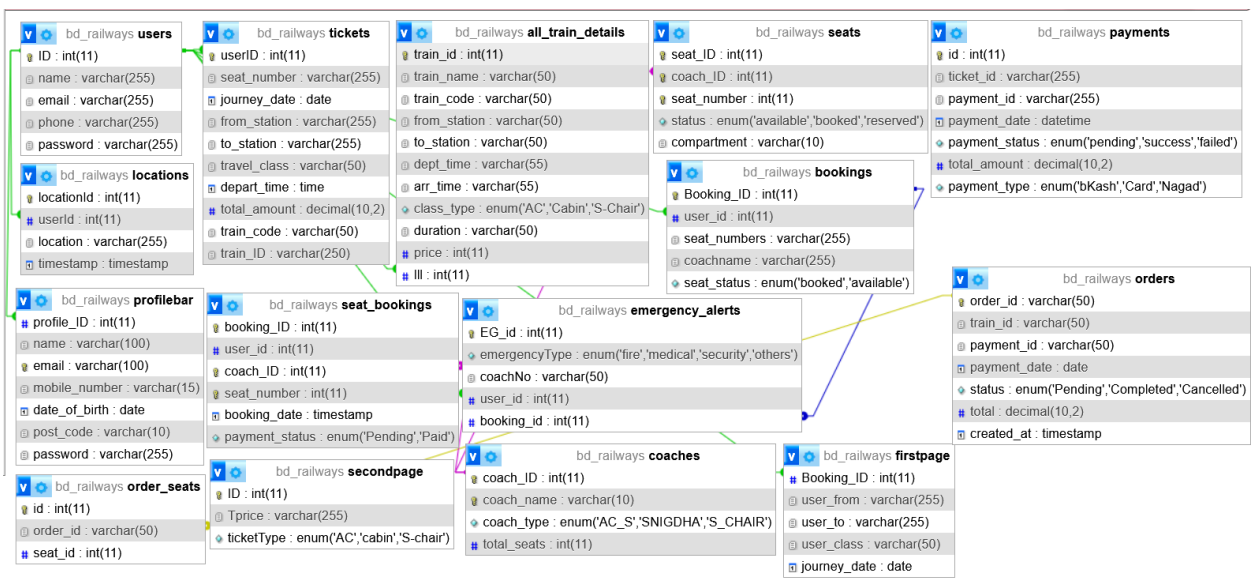
## ERD & Schema Diagram

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## DDL to Create Database

CREATE DATABASE bd_railways;

USE bd_railways;

CREATE TABLE Users (

user_id INT PRIMARY KEY AUTO_INCREMENT,

```
name VARCHAR(255),
email VARCHAR(255) UNIQUE,
password VARCHAR(255),
role ENUM('passenger', 'admin')
);
CREATE TABLE Trains (
train_id INT PRIMARY KEY AUTO_INCREMENT,
train_name VARCHAR(255),
source VARCHAR(255),
destination VARCHAR(255),
departure_time DATETIME,
arrival_time DATETIME
);
CREATE TABLE Bookings (
booking_id INT PRIMARY KEY AUTO_INCREMENT,
user_id INT,
train_id INT,
seat_number VARCHAR(10),
status ENUM('confirmed', 'cancelled'),
FOREIGN KEY (user_id) REFERENCES Users(user_id),
FOREIGN KEY (train_id) REFERENCES Trains(train_id)
);
CREATE TABLE `admin_train` (
`id` int(11) NOT NULL,
`train_name` varchar(255) NOT NULL,
`train_code` varchar(255) NOT NULL,
```

```
`from_station` varchar(255) NOT NULL,
`to_station` varchar(255) NOT NULL,
`dept_time` time NOT NULL,
`arr_time` time NOT NULL,
`class_type` varchar(255) NOT NULL,
`duration` varchar(255) NOT NULL,
`price` decimal(10,2) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci
COMMENT='ADMIN Train Details';
CREATE TABLE `all_train_details` (
`train_id` int(11) NOT NULL,
`train_name` varchar(50) NOT NULL,
`train_code` varchar(50) NOT NULL,
`from_station` varchar(50) NOT NULL,
`to_station` varchar(50) NOT NULL,
`dept_time` varchar(55) NOT NULL,
`arr_time` varchar(55) NOT NULL,
`class_type` enum('AC','Cabin','S-Chair') NOT NULL,
`duration` varchar(50) NOT NULL,
`price` int(11) NOT NULL,
`lll` int(11) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci
COMMENT='All_Details (About Train)';
CREATE TABLE `bookings` (
`Booking_ID` int(11) NOT NULL,
`user_id` int(11) NOT NULL,
`coachname` varchar(10) DEFAULT NULL,
```

```
`seat_status` varchar(20) DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
CREATE TABLE `coaches` (
`coach_ID` int(11) NOT NULL,
`coach_name` varchar(10) NOT NULL,
`coach_type` enum('AC_S','SNIGDHA','S_CHAIR') NOT NULL,
`total_seats` int(11) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
CREATE TABLE `emergency_alerts` (
`EG_id` int(11) NOT NULL,
`emergencyType` enum('fire','medical','security','others') NOT NULL,
`coachNo` varchar(50) NOT NULL,
`user_id` int(11) NOT NULL,
`booking_id` int(11) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
CREATE TABLE `firstpage` (
`Booking_ID` int(11) NOT NULL COMMENT 'Primary Key',
`user_from` varchar(255) DEFAULT NULL COMMENT 'User From',
`user_to` varchar(255) DEFAULT NULL COMMENT 'User To',
`user_class` varchar(50) DEFAULT NULL COMMENT 'User Class',
`journey_date` date DEFAULT NULL COMMENT 'Journey Date'
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
CREATE TABLE `locations` (
`locationId` int(11) NOT NULL,
`userId` int(11) DEFAULT NULL,
`location` varchar(255) DEFAULT NULL,
```

```
`timestamp` timestamp NOT NULL DEFAULT current_timestamp()
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
CREATE TABLE `orders` (
`order_id` varchar(50) NOT NULL,
`train_id` varchar(50) NOT NULL,
`payment_id` varchar(50) NOT NULL,
`payment_date` date NOT NULL,
`status` enum('Pending','Completed','Cancelled') NOT NULL,
`total` decimal(10,2) NOT NULL,
`created_at` timestamp NOT NULL DEFAULT current_timestamp()
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
CREATE TABLE `order_seats` (
`id` int(11) NOT NULL,
`order_id` varchar(50) NOT NULL,
`seat_id` int(11) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
CREATE TABLE `payments` (
`id` int(11) NOT NULL,
`ticket_id` varchar(255) NOT NULL,
`payment_id` varchar(255) NOT NULL,
`payment_date` datetime NOT NULL,
`payment_status` enum('pending','success','failed') NOT NULL,
`total_amount` decimal(10,2) NOT NULL,
`payment_type` enum('bKash','Card','Nagad') NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
CREATE TABLE `seats` (
```

```
`Seat_ID` int(11) NOT NULL,
`Booking_ID` int(11) NOT NULL,
`seat_numbers` int(11) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
CREATE TABLE `secondpage` (
`ID` int(11) NOT NULL,
`Tprice` varchar(255) DEFAULT NULL COMMENT 'Seat,s Price',
`ticketType` enum('AC','cabin','S-chair') DEFAULT NULL COMMENT 'Ticket Type'
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
CREATE TABLE `seat_bookings` (
`booking_ID` int(11) NOT NULL,
`user_id` int(11) NOT NULL,
`coach_ID` int(11) NOT NULL,
`seat_number` int(11) NOT NULL,
`booking_date` timestamp NOT NULL DEFAULT current_timestamp(),
`payment_status` enum('Pending','Paid') DEFAULT 'Pending'
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
CREATE TABLE `tickets` (
`userID` int(11) NOT NULL,
`seat_number` varchar(255) NOT NULL,
`journey_date` date NOT NULL,
`from_station` varchar(255) NOT NULL,
`to_station` varchar(255) NOT NULL,
`travel_class` varchar(50) NOT NULL,
`depart_time` time NOT NULL,
`total_amount` decimal(10,2) NOT NULL,
```

```
`train_code` varchar(50) NOT NULL,
`train_ID` varchar(250) DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
CREATE TABLE `users` (
`ID` int(11) NOT NULL,
`name` varchar(255) DEFAULT NULL,
`email` varchar(255) DEFAULT NULL,
`phone` varchar(255) DEFAULT NULL,
`password` varchar(255) DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

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**DML Used in the Application**

```
ALTER TABLE `admin_train`
  ADD PRIMARY KEY (`id`);
ALTER TABLE `all_train_details`
  ADD PRIMARY KEY (`train_id`),
  ADD KEY `fk_UID` (`lll`);
ALTER TABLE `bookings`
  ADD PRIMARY KEY (`Booking_ID`);
ALTER TABLE `coaches`
  ADD PRIMARY KEY (`coach_ID`),
  ADD UNIQUE KEY `coach_name` (`coach_name`);
ALTER TABLE `emergency_alerts`
  ADD PRIMARY KEY (`EG_id`),
  ADD KEY `user_id` (`user_id`),
```

```
  ADD KEY `booking_id` (`booking_id`);
ALTER TABLE `locations`
  ADD PRIMARY KEY (`locationId`),
  ADD KEY `userId` (`userId`);
ALTER TABLE `orders`
  ADD PRIMARY KEY (`order_id`);
ALTER TABLE `order_seats`
  ADD PRIMARY KEY (`id`),
  ADD KEY `order_id` (`order_id`),
  ADD KEY `seat_id` (`seat_id`);
ALTER TABLE `payments`
  ADD PRIMARY KEY (`id`);
ALTER TABLE `profilebar`
  ADD UNIQUE KEY `email` (`email`),
  ADD KEY `profile_ID` (`profile_ID`);
ALTER TABLE `seats`
  ADD PRIMARY KEY (`Seat_ID`),
  ADD UNIQUE KEY `unique_seat` (`Booking_ID`,`seat_numbers`);
ALTER TABLE `seat_bookings`
  ADD PRIMARY KEY (`booking_ID`),
  ADD UNIQUE KEY `unique_booking` (`coach_ID`,`seat_number`);
ALTER TABLE `secondpage`
  ADD PRIMARY KEY (`ID`);
ALTER TABLE `tickets`
  ADD PRIMARY KEY (`userID`);
```

```
ALTER TABLE `users`
ADD PRIMARY KEY (`ID`);
ALTER TABLE `admin_train`
 MODIFY `id` int(11) NOT NULL AUTO_INCREMENT;
ALTER TABLE `all_train_details`
  MODIFY `train_id` int(11) NOT NULL AUTO_INCREMENT, AUTO_INCREMENT;
ALTER TABLE `bookings`
  MODIFY `Booking_ID` int(11) NOT NULL AUTO_INCREMENT, AUTO_INCREMENT;
ALTER TABLE `coaches`
  MODIFY `coach_ID` int(11) NOT NULL AUTO_INCREMENT, AUTO_INCREMENT;
ALTER TABLE `locations`
  MODIFY `locationId` int(11) NOT NULL AUTO_INCREMENT;
ALTER TABLE `order_seats`
  MODIFY `id` int(11) NOT NULL AUTO_INCREMENT, AUTO_INCREMENT;
ALTER TABLE `payments`
  MODIFY `id` int(11) NOT NULL AUTO_INCREMENT, AUTO_INCREMENT;
ALTER TABLE `seats`
  MODIFY `Seat_ID` int(11) NOT NULL AUTO_INCREMENT, AUTO_INCREMENT;
ALTER TABLE `seat_bookings`
  MODIFY `booking_ID` int(11) NOT NULL AUTO_INCREMENT;
ALTER TABLE `secondpage`
  MODIFY `ID` int(11) NOT NULL AUTO_INCREMENT, AUTO_INCREMENT;
ALTER TABLE `users`
  MODIFY `ID` int(11) NOT NULL AUTO_INCREMENT, AUTO_INCREMENT;
ALTER TABLE `all_train_details`
  ADD CONSTRAINT `fk_UID` FOREIGN KEY (`lll`) REFERENCES `users` (`ID`);
ALTER TABLE `emergency_alerts`
```

```
  ADD CONSTRAINT `emergency_alerts_ibfk_1` FOREIGN KEY (`user_id`) REFERENCES `users` (`ID`),

  ADD CONSTRAINT `emergency_alerts_ibfk_2` FOREIGN KEY (`booking_id`) REFERENCES `bookings_old` (`Booking_ID`);

ALTER TABLE `firstpage`

  ADD CONSTRAINT `firstpage_ibfk_1` FOREIGN KEY (`Booking_ID`) REFERENCES `users` (`ID`);

ALTER TABLE `locations`

  ADD CONSTRAINT `locations_ibfk_1` FOREIGN KEY (`userId`) REFERENCES `users` (`ID`);
ALTER TABLE `order_seats`
 ADD CONSTRAINT `order_seats_ibfk_1` FOREIGN KEY (`order_id`) REFERENCES `orders` (`order_id`),
 ADD CONSTRAINT `order_seats_ibfk_2` FOREIGN KEY (`seat_id`) REFERENCES `seats` (`seat_id`);

ALTER TABLE `profilebar`

  ADD CONSTRAINT `profilebar_ibfk_1` FOREIGN KEY (`profile_ID`) REFERENCES `users` (`ID`);

ALTER TABLE `seats`

  ADD CONSTRAINT `seats_ibfk_1` FOREIGN KEY (`Booking_ID`) REFERENCES `bookings` (`Booking_ID`);

ALTER TABLE `seat_bookings`

  ADD CONSTRAINT `seat_bookings_ibfk_1` FOREIGN KEY (`coach_ID`) REFERENCES `coaches` (`coach_ID`);

ALTER TABLE `tickets`

  ADD CONSTRAINT `tickets_ibfk_1` FOREIGN KEY (`userID`) REFERENCES `users` (`ID`),

  ADD CONSTRAINT `tickets_ibfk_2` FOREIGN KEY (`train_id`) REFERENCES `trains` (`train_id`);
```

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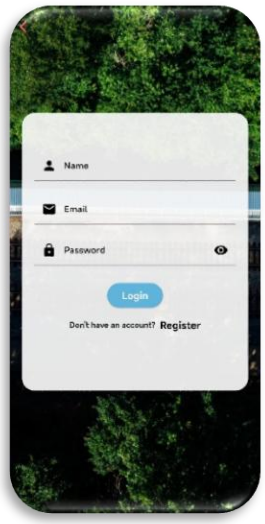
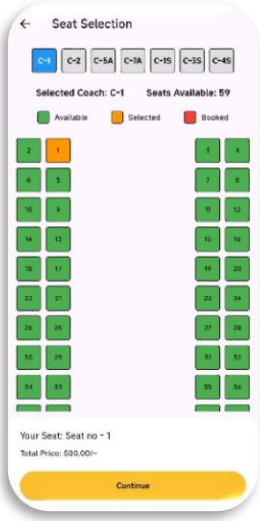
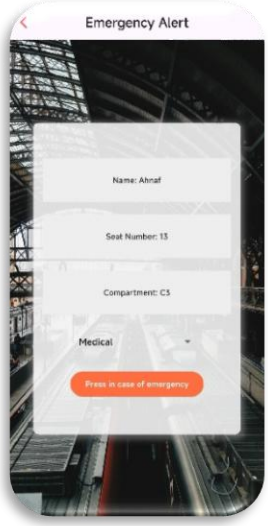
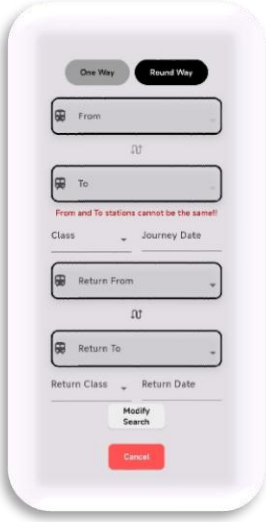
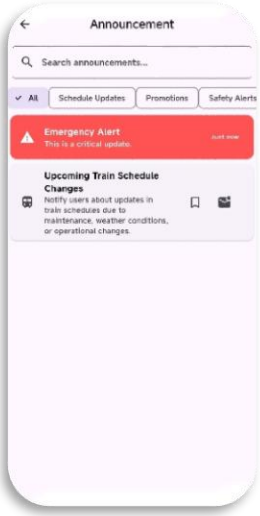
## Screenshots of the Application

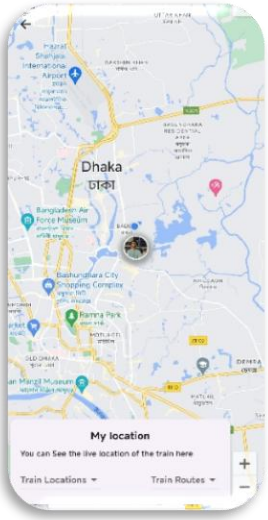




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## Challenges Faced & How We Overcame Them

1. Database adaptation

Challenge: Design a customized relationship database.

Solution: Generalization techniques and sequencing for efficiency.

2. Real -time train tracking

Challenge: Effective live location handling updates while avoiding $ 50 Google Console fee.

Solution: To track train sites without excessive cost, detection of alternatives such as OpenStreetmap and Firebase RealTime database.

3. Secure payment integration

Challenge: Ensure a safe and easy payment process.

Solution: Stripe API was implemented for reliable transactions.

4. Change buttons and information performance

Challenge: Securing the revised button updates the user interface correctly and shows accurate information.

Solution: UI updates were implemented dynamically and to handle data recovery effectively to get the GETX state management.

5. Limited scheduled flexibility

Challenge: The app was originally limited to only six fixed routes.

Solution: To support a dynamic routing system, the database and app logic are redesigned, which allows easy expansion of routes to the future.